QuantX × C12 hackathon

Automating the calibration of a double quantum dot

I. Interdot detection · II. Contrast optimisation

Team 1 · 2 October 2026

OUTLINE

Presentation outline

I

Interdot detection

1. An algorithmic approach: line filters, no training
2. Detection using a U-Net, and its robustness to lab artefacts

II

Contrast optimisation

Our optimizer based on the constant interaction model: one translation for all sticks, virtual gates, sparse exploration of the barrier cube, local refinement

I

Interdot detection

1. An algorithmic approach · 2. Detection using a U-Net

Image processing to detect the interdots

Raw image

Binary mask

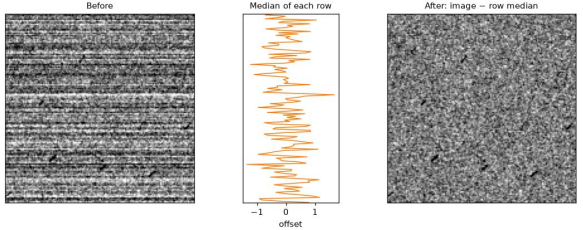
1 Row median
image − median of each row

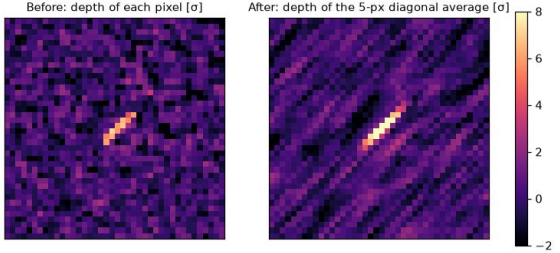
2 Robust depth
d = (median − a) / 1.4826 ·MAD

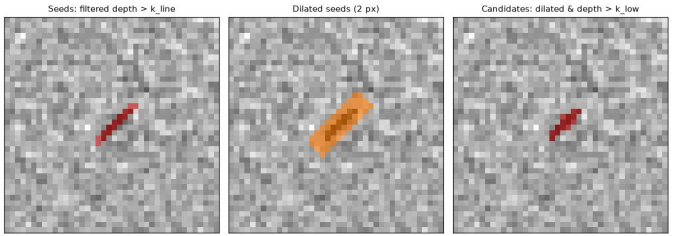
3 Line filter
5-px average at 45°, seeds, candidates

4 Half-max cut
keep d > 0.6·A per component

5 Orientation
8 line directions, keep 45° ± 22.5°







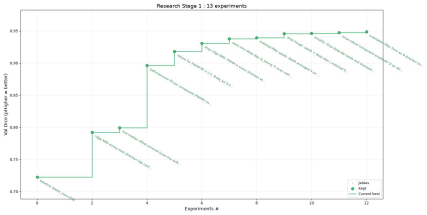


Image processing to detect the interdots

Raw image

Binary mask

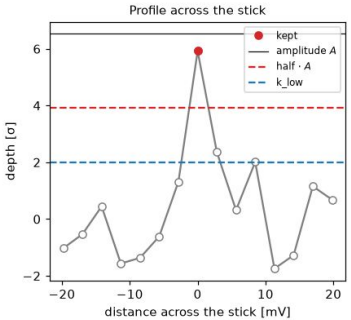
1 Row median
image − median of each row

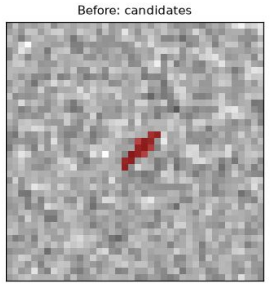
2 Robust depth
d = (median − a) / 1.4826 ·MAD

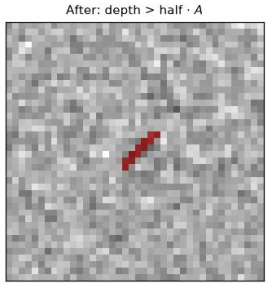
3 Line filter
5-px average at 45°, seeds, candidates

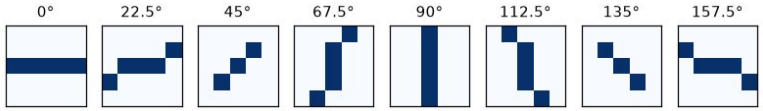
4 Half-max cut
keep d > 0.6·A per component

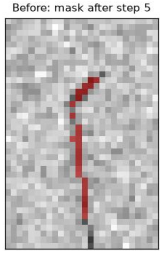
5 Orientation
8 line directions, keep 45° ± 22.5°

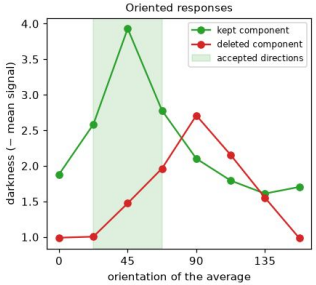


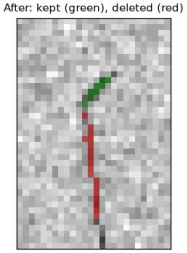




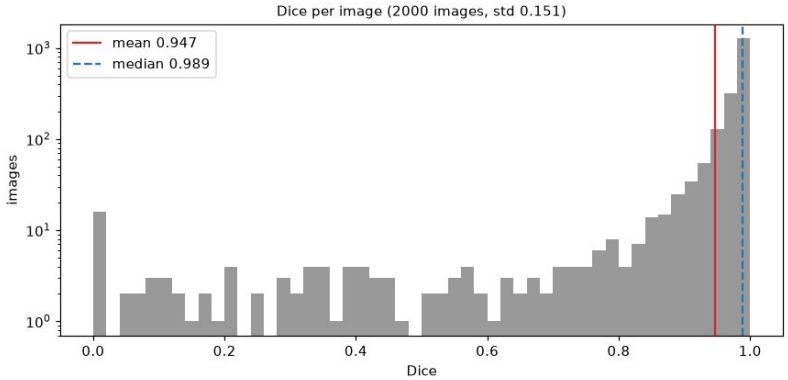








Results

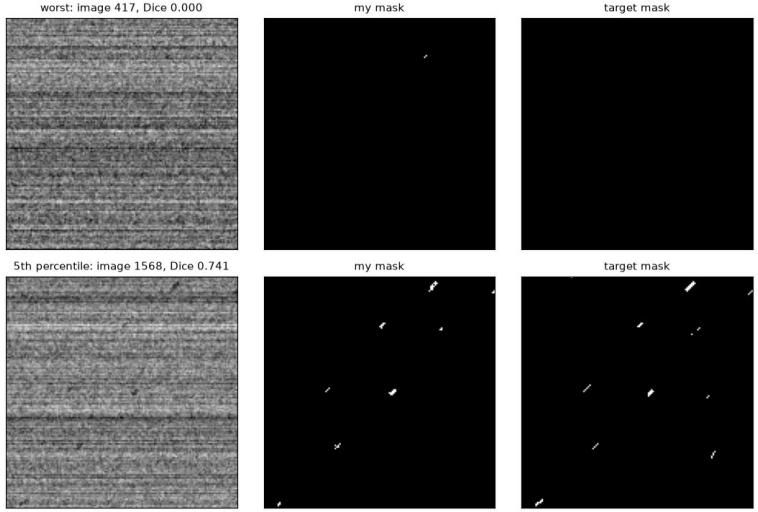


0.947
mean Dice (std 0.151)

0.989
median Dice

90.9 %
images with Dice ≥ 0.90

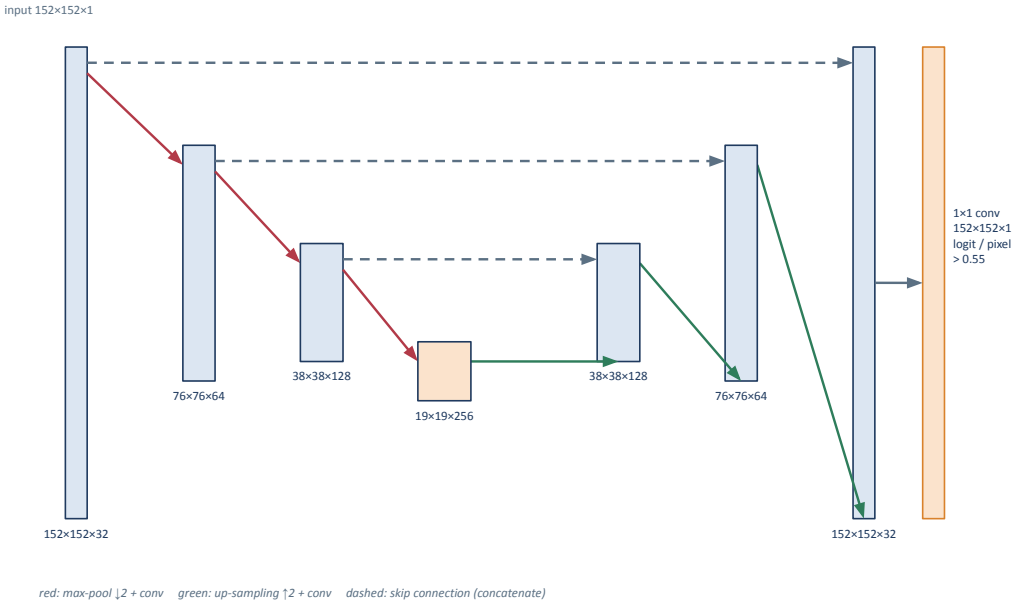
0.741
5th percentile



Limitations

- Stick direction 45° hard-coded in the line filter and orientation filter
- Sticks below 3.5 σ after the 5-px average are lost; sticks shorter than 5 px gain less

UNet : Deep Learning architecture



0.980

object F1, 200 clean test scenes (precision 0.996, recall 0.965)

0.990

tolerant F1 (±1 px)

0.984

strict pixel Dice

Robustness suite: 9 families × up to 3 levels = 25 sets (white and 1/f noise, drift, charge jumps, stripes, lock-in low-pass, saturation, outlier pixels, inverted polarity).

0.872

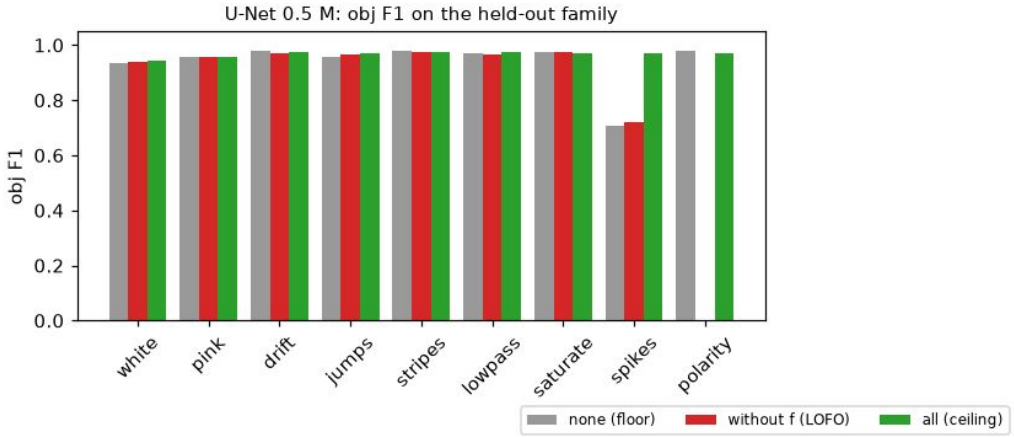
robustness score = mean of obj F1(perturbed) / obj F1(clean) over the 25 sets

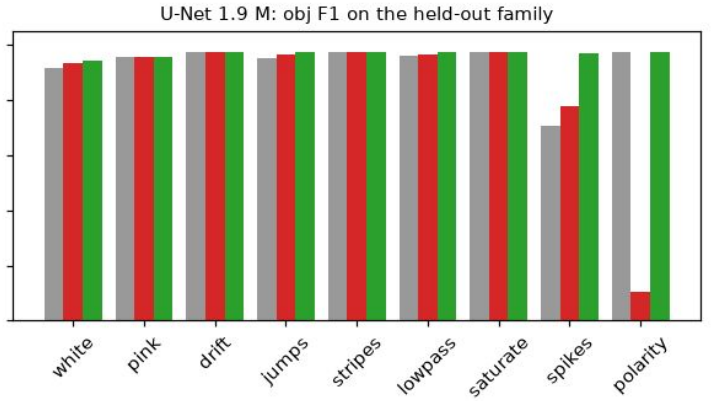
Preprocessing + randomised artefacts, tested leave-one-family-out

Physical preprocessing
row median removal · MAD rescaling · despiking: a pixel > 6 σ whose 8 neighbours stay weak → its 3×3 median

Randomised training
8 artefact families, each with probability 0.2 and uniform severity; polarity augmented

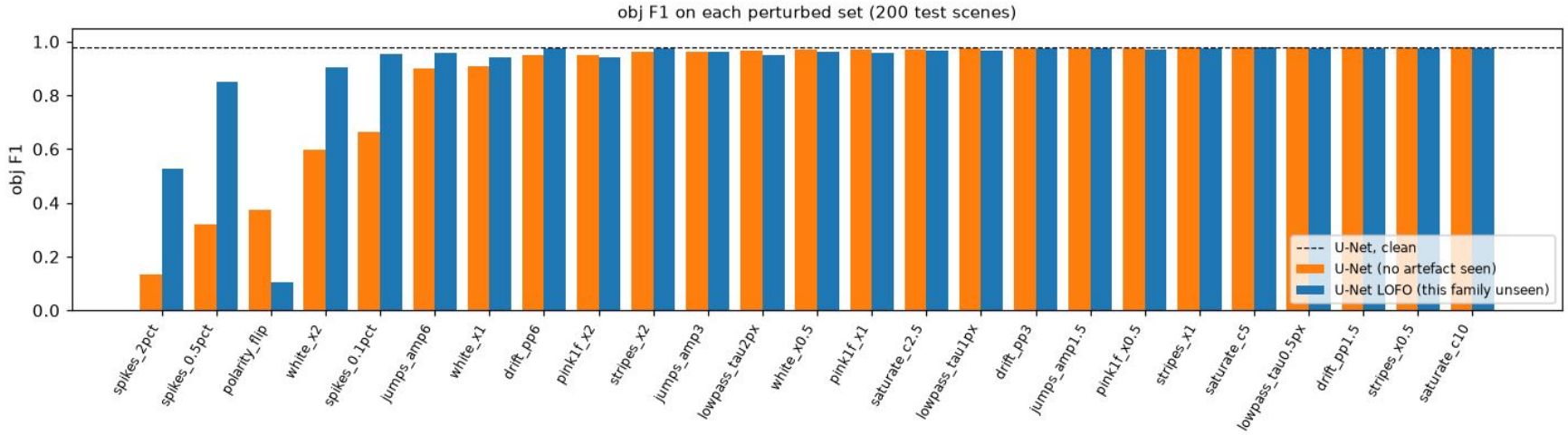
Run	Families seen	Role
none	none	floor
no-f	**all but f**	**f unknown: the test**
all	all, f included	ceiling





Object F1 on the held-out family, 2 capacities × 11 runs (8,000 steps each)

U-Net vs U-Net LOFO (Leave One Family Out)



	U-Net	U-Net LOFO
Object F1, clean test	**0.980**	0.975
Object F1, stage-2 frames	**0.964**	0.951
Robustness, 25 sets	0.872	**0.930**
Robustness without polarity	0.892	**0.964**

II

Our optimizer

Based on the constant interaction model

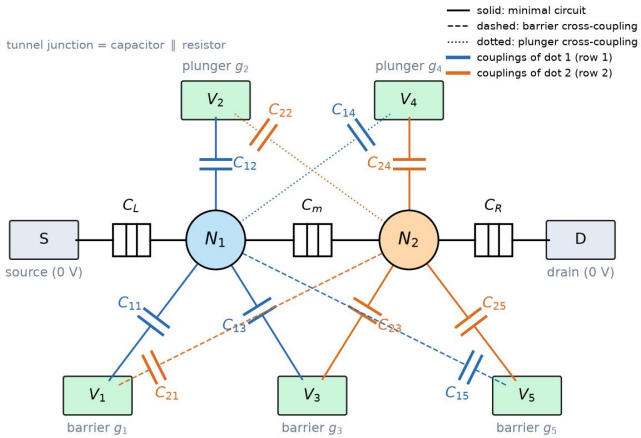
The translation model

- Stick between (N_1+1, N_2) and (N_1, N_2+1): F equal $\Rightarrow u_1 - u_2 = \text{const}$
- Its end points (triple points) also fix $u_1, u_2 \Rightarrow$ a point of a stick is a fixed u^*
- Barriers move by δb; stay on u^*: $\alpha_P\, \delta p + \alpha_B\, \delta b = 0$

$$\delta p = D\, \delta b, \quad D = -\alpha_P^{-1}\, \alpha_B = -C_gP^{-1}\, C_gB$$

$$t_x(b) = \Sigma_k\, D_{xk}\, b_k + \Sigma_{k,l}\, Q_{x,kl}\, b_k\, b_l\,, \quad x \in \{g2, g4\}, \quad Q_x \text{ symmetric}$$

	Content	Unknowns
Linear D	2 × 3 lever arms	6
Quadratic Q_{g2}, Q_{g4}	2 symmetric 3 × 3 matrices	12
Total		**18**



Charges

$Q = C\,\varphi - C_g\, v, \quad C = [[C_1, -C_m], [-C_m, C_2]]$

Potentials

$\varphi = C^{-1}\, Q + \alpha\, v, \quad \alpha = C^{-1}\, C_g$ (lever arms)

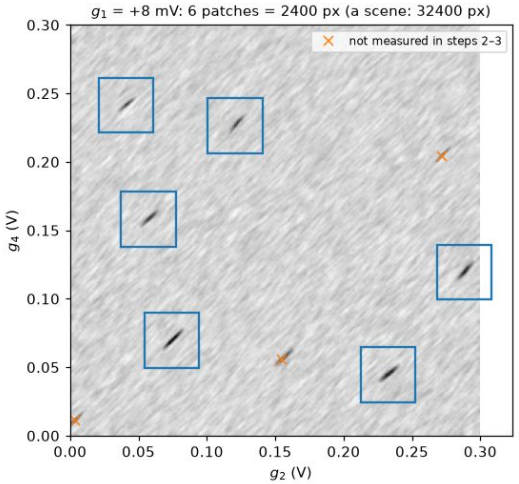
Free energy ($Q = -|e|\, N$)

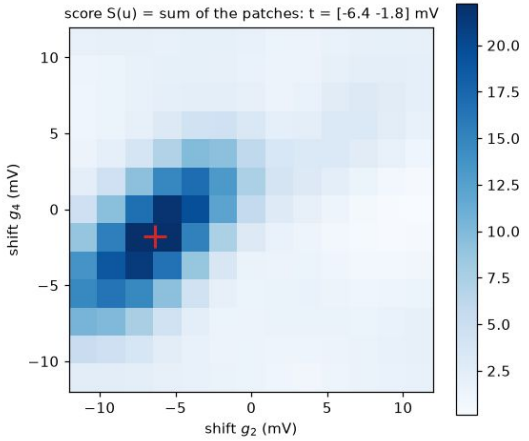
$F(N) = \tfrac{1}{2}\, N^T\, E\, N - N \cdot u, \quad E = e^2\, C^{-1}$

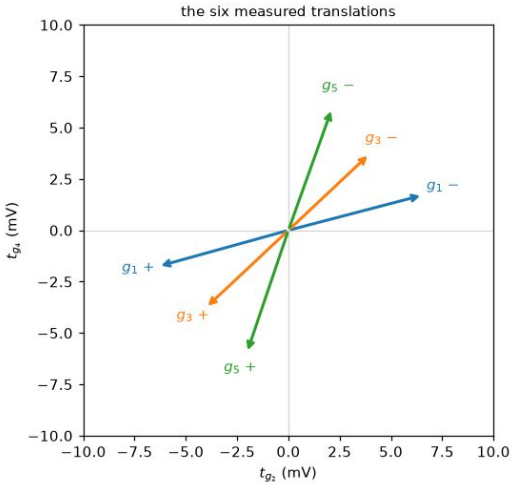
Voltages enter only through

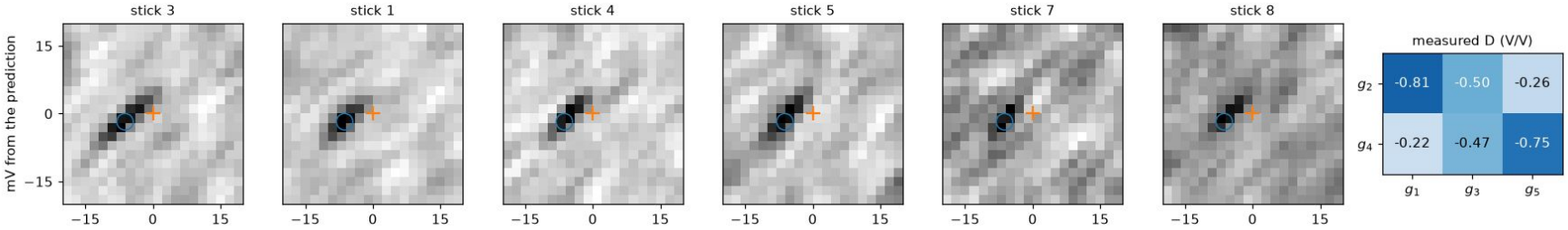
$u = |e|\, \alpha\, v$ (linear in v)

Calibration of the model D from ±8 mV and Q +- 120mV

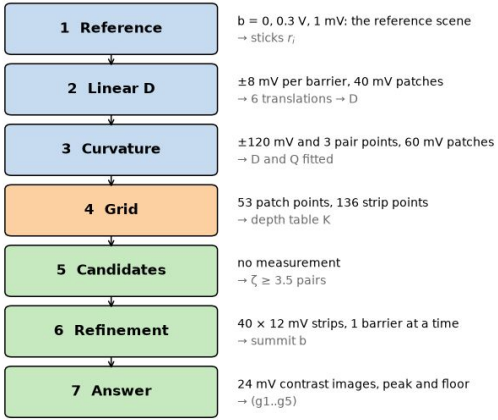


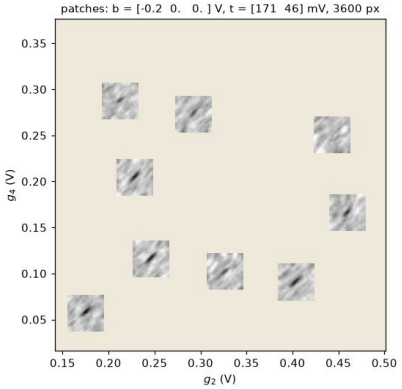


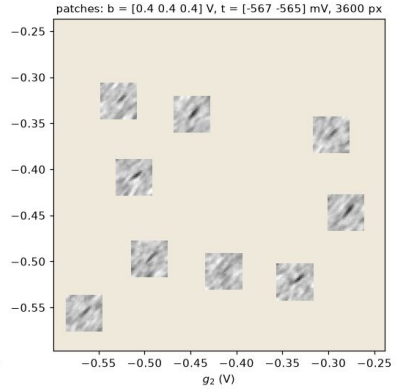


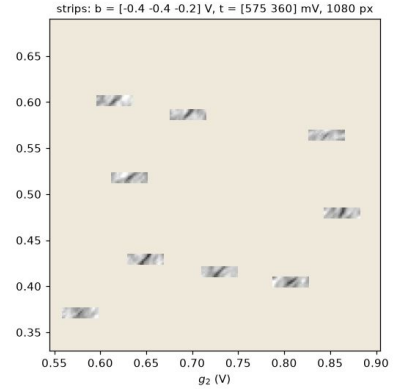


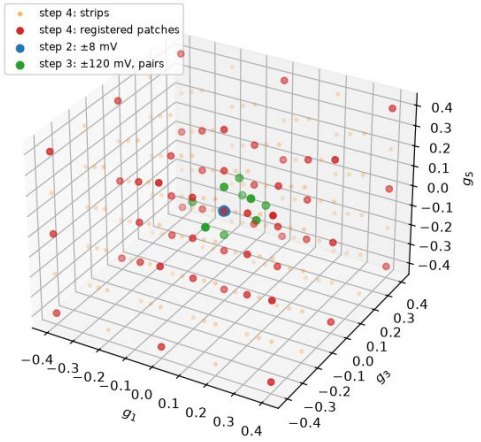
The procedure







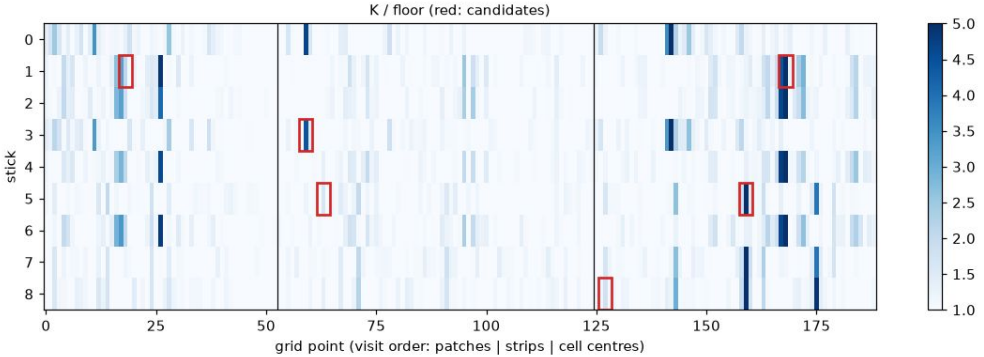


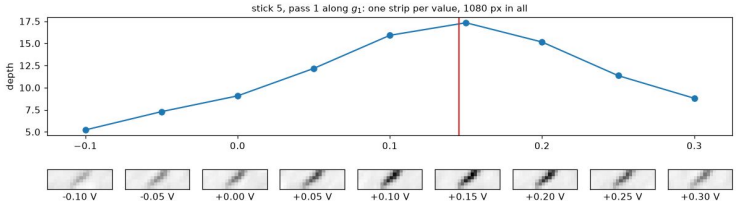


	Window	Size, step	Pixels	Used for
A	Reference scene	300 mV, 1 mV	90,000	step 1
B	Scene	360 mV, 2 mV	32,400	fallback only
C	Wide patch	60 mV, 2 mV	900	steps 3–4 (young model)
D	Patch	40 mV, 2 mV	400	steps 2, 4, 7: registration
E	Strip	40 × 12 mV, 2 mV	120	steps 4, 6: depth only
F	Contrast image	24 mV, 0.5 mV	2,304	step 7: absolute contrast

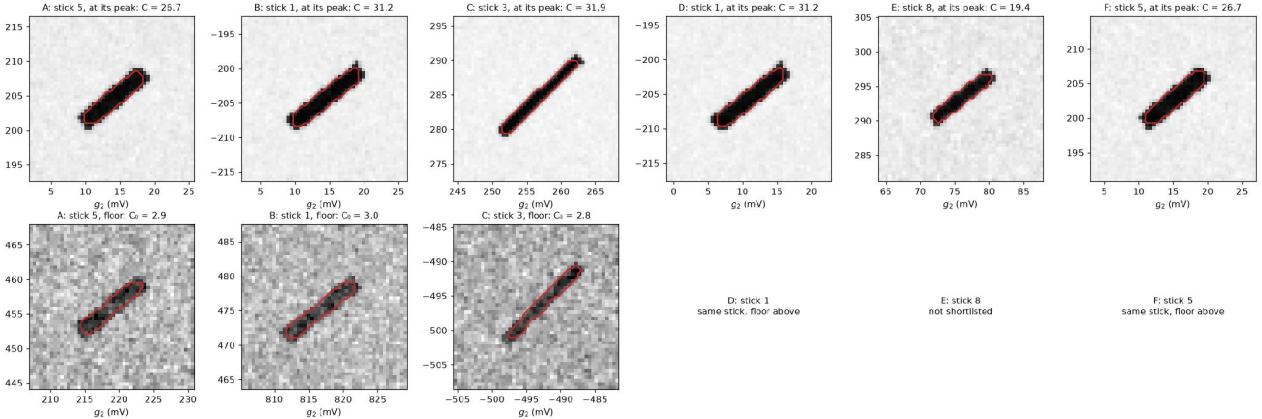
Candidates and refinement

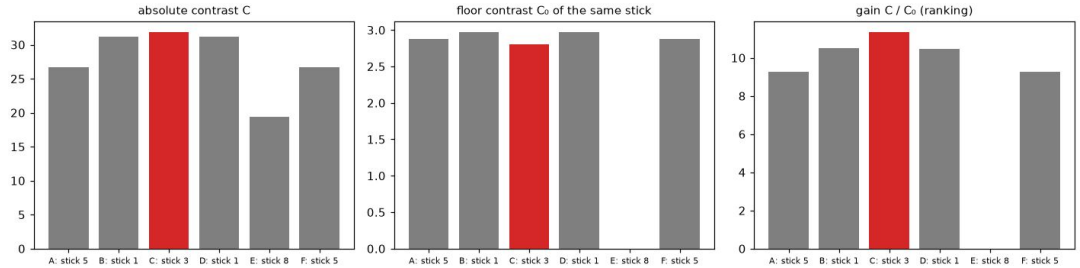
$$\zeta_{m,i} = (K_{m,i} - f_i) / s_i, \quad f_i = \text{median}_m\, K_{m,i}, \quad s_i = 1.4826 \cdot \text{MAD}_m\, K_{m,i}$$





Absolute contrast, then gain against the stick's floor

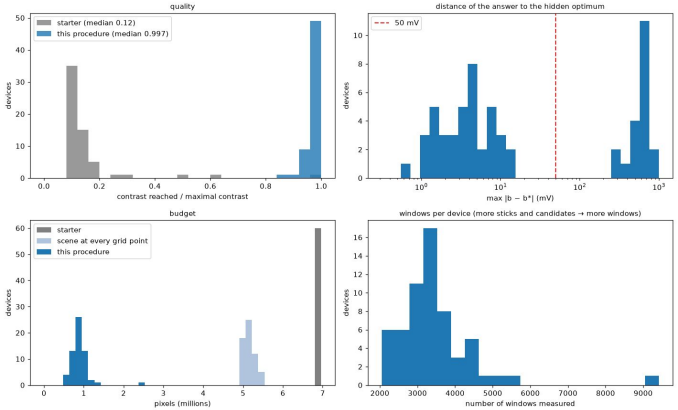




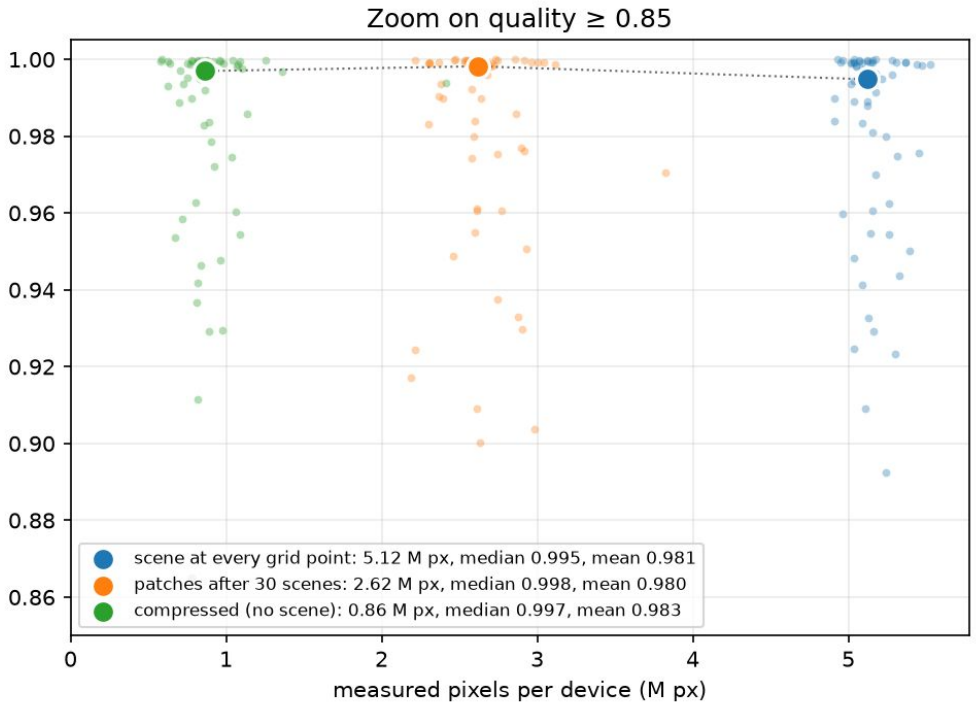
- C: 24 mV image at 0.5 mV, rows corrected; core = connected pixels below half of the smoothed minimum, eroded by 1 px; C = −mean(core), averaged over 4 images
- Each stick has its own intensity ⟶ a bright stick on a lower peak can beat a dim stick on the highest
- Candidates with C ≥ 0.75 · max C: floor C_0 of the same stick at the opposite corner of the cube
- Ranking by gain C / C_0

Device 1007: candidate C (stick 3), C = 31.9, C_0 = 2.8, gain ≈ 11.4

Results



	Ours	Scenes	Starter
Median quality	**0.997**	0.995	0.116
Mean quality	**0.983**	0.981	0.154
Quality ≥ 0.95	**87 %**	85 %	2 %
Min quality	0.865	0.892	0.095
Optimal peak refined	**98 %**	93 %	—
Median pixels	**0.86 M**	5.12 M	6.86 M



Conclusion

Detection

Line filters along the stick direction: mean Dice 0.947 with no training. U-Net: object F1 0.980; with PhysInput it keeps 0.975 and raises robustness from 0.872 to 0.930 on unseen artefacts.

Optimisation

Constant interaction model ⇒ one translation D b + Q(b, b) for all sticks ⇒ measure only small windows where the sticks are predicted.

Result

Median quality 0.997 with 0.86 M px: 8× fewer pixels than the starter, which reaches 0.116.

Thank you — questions?